

CS464

Week 03

Classical Encryption Techniques

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Introduction to Classical Encryption Techniques

Classical encryption techniques are examined in this section to illustrate the basic approaches to symmetric encryption used today and the types of cryptanalytic attacks that must be anticipated in modern cryptographic systems.

Substitution

The first basic building block where plaintext letters are replaced by other letters, numbers, or symbols.

Transposition

The second basic building block where plaintext letters are permuted or rearranged in position.

 **Combined Systems:** Advanced encryption systems combine both substitution and transposition techniques to achieve stronger security.

Substitution Techniques - Overview

Definition of Substitution Technique

A **substitution technique** is one in which the letters of plaintext are replaced by other letters or by numbers or symbols, creating a fundamental transformation of the original message content.

A Letter Substitution

Plaintext letters are replaced with other letters from the same or extended alphabet.

Number/Symbol Substitution

Letters can be replaced with numerical values or special symbols for added complexity.

Bit-Level Perspective

If the plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns, which is how modern computers implement these classical techniques.

Caesar Cipher - Introduction



Historical Origin

The **Caesar cipher** represents the earliest known and simplest use of a substitution cipher, employed by Julius Caesar for secure military communications.

How It Works

Each letter of the alphabet is replaced with the letter standing **three places further down** the alphabet.

$A \rightarrow D, B \rightarrow E, C \rightarrow F, \dots$

Alphabet Wrapping

The alphabet **wraps around**, meaning the letter following Z is A, creating a circular substitution pattern.

$X \rightarrow A, Y \rightarrow B, Z \rightarrow C$

i Key Characteristic: The Caesar cipher uses a fixed shift of 3 positions, making it a monoalphabetic substitution cipher with a very small key space of only 25 possible variations.

Caesar Cipher - Example

Encryption Example

Plaintext:

meet me after the toga party

Ciphertext:

PHHW PH DIWHU WKH WRJD SDUWB

Complete Transformation Table

Plain alphabet:

a b c d e f g h i j k l m
n o p q r s t u v w x y z

Cipher alphabet:

D E F G H I J K L M N O P
Q R S T U V W X Y Z A B C

m (12) →

P (15)

e (4) →

H (7)

t (19) →

W (22)

Caesar Cipher - Numerical Representation

To perform mathematical operations on letters, we assign a **numerical equivalent** to each letter of the alphabet, creating a mapping that enables arithmetic encryption algorithms.

Letter-to-Number Mapping

a 0	b 1	c 2	d 3	e 4	f 5	g 6	h 7	i 8	j 9	k 10	l 11	m 12
n 13	o 14	p 15	q 16	r 17	s 18	t 19	u 20	v 21	w 22	x 23	y 24	z 25

📌 **Convention:** Plaintext is always in lowercase, ciphertext is in uppercase, and key values are in italicized lowercase, as used throughout this course material.

Caesar Cipher - Mathematical Formulation

Encryption Algorithm

For each plaintext letter p , the ciphertext letter C is computed using modular arithmetic:

$$C = E(3, p) = (p + 3) \bmod 26$$

General Caesar Algorithm

A shift may be of any amount from 1 to 25:

$$C = E(k, p) = (p + k) \bmod 26$$

where $k \in \{1, 2, 3, \dots, 25\}$

Decryption Algorithm

To recover plaintext from ciphertext:

$$p = D(k, C) = (C - k) \bmod 26$$

Same key k used for decryption

Example Calculation

Encrypt 'm' (12):
 $(12 + 3) \bmod 26 = 15 \rightarrow \text{'P'}$



Decrypt 'P' (15):
 $(15 - 3) \bmod 26 = 12 \rightarrow \text{'m'}$

Caesar Cipher - Brute-Force Cryptanalysis

Vulnerability to Brute-Force Attack

If it is known that a given ciphertext is a Caesar cipher, then a **brute-force cryptanalysis** is easily performed by simply trying all the 25 possible keys.



25 Possible Keys

Very small key space makes exhaustive search feasible.



Plaintext Recognition

The correct plaintext "leaps out" as the readable message.



Rapid Break

Attack completes quickly due to minimal key space.

Attack Process

- 1 Try key $k=1$: shift all letters back by 1 position.
- 2 Try key $k=2$: shift all letters back by 2 positions.
- 3 Continue until $k=25$, checking each output for meaningful text.

Brute-Force Cryptanalysis of Caesar Cipher

	PHHW	PH	DIWHU	WKH	WRJD	SDUWB
KEY						
1	oggv	og	chvgt	vjg	vqic	rctva
2	nffu	nf	bgufs	uif	uphb	qbsuz
3	meet	me	after	the	toga	party
4	ldds	ld	zesdq	sgd	snfz	ozqsx
5	kccr	kc	ydrpc	rfe	rmey	nyprw
6	jbbq	jb	xcqbo	qeb	qldx	mxoqv
7	iaap	ia	wbpan	pda	pkcw	lwnpu
8	hzzo	hz	vaozm	ocz	ojbv	kvmot
9	gyyn	gy	uznyl	nby	niau	julns
10	fxxm	fx	tymxk	max	mhzt	itkmr
11	ewwl	ew	sxlwj	lzw	lgys	hsjlg
12	dvvk	dv	rwkvi	kyv	kfxr	grikp
13	cuuj	cu	qvjuh	jxu	jewq	fqhjo
14	btti	bt	putg	iwt	idvp	epgin
15	assh	as	othsf	hvs	hcuo	dofhm
16	zrrg	zr	nsgre	gur	gbtn	cnegl
17	yqqf	yq	mrfqd	ftq	fasm	bmdfk
18	xppe	xp	lqepc	esp	ezrl	alcej
19	wood	wo	kpdob	dro	dyqk	zkbdi
20	vnnv	vn	jocna	cqn	cxpj	yjach
21	ummb	um	inbmz	bpm	bwoi	xizbg
22	tlla	tl	hmaly	aol	avnh	whyaf
23	skkz	sk	glzkx	znk	zumg	vgxze
24	rjjy	rj	fkyjw	ymj	ytlf	ufwyd
25	qiix	qi	ejxiv	xli	xske	tevxc

Three Characteristics Enabling Brute-Force Attack

Three important characteristics of the Caesar cipher problem enable the successful use of brute-force cryptanalysis:

1

Known Algorithms

The encryption and decryption algorithms are known. In most networking situations, we can assume that the algorithms are publicly known, which is a standard assumption in modern cryptography known as Kerckhoffs's principle.

2

Small Key Space

There are only 25 keys to try, making exhaustive search computationally trivial. What generally makes brute-force cryptanalysis impractical is using an algorithm that employs a large number of keys.

3

Recognizable Language

The language of the plaintext is known and easily recognizable. When the correct key is applied, the resulting plaintext is immediately identifiable as valid language.

Practical Considerations for Brute-Force Attacks

In most networking situations, we can assume that the **algorithms are known**. What makes brute-force cryptanalysis impractical is using an algorithm with a large key space.



Caesar Cipher

Key Size: 5 bits (25 keys)

Key Space: 25

Security: Trivial



Triple DES

Key Size: 168 bits

Key Space: 2^{168}

Approximate: $> 3.7 \times 10^{50}$



Key Insight: The triple DES algorithm, examined in Chapter 7, makes brute-force attacks computationally infeasible due to its enormous key space of greater than 3.7×10^{50} possible keys.

Language Recognition Challenges

The **third characteristic** enabling brute-force attacks - recognizable plaintext language - can be undermined through various techniques that obscure the statistical patterns of natural language.

Unknown Language

If the language of the plaintext is unknown, then plaintext output may not be recognizable even with the correct decryption key.

Compression

The input may be compressed, removing the statistical patterns that make natural language recognizable.

Impact on Cryptanalysis

As shown in Figure ahead, if a compressed file is encrypted with a simple substitution cipher (expanded to include more than just 26 alphabetic characters), the plaintext may not be recognized when it is uncovered during brute-force cryptanalysis.

 **Defense Strategy:** Compression before encryption can make ciphertext-only attacks more difficult by removing the language statistics that cryptanalysts rely upon.

Sample of Compressed Text

~+Wμ"— Ω-O)≤4{∞‡, ë~Ω%ràu·-Í ◇-Z-
Ú≠2Ò#Åæð æ«q7,Ωn·®3NÔÚ ℰz'Y-f∞Í[±Û_ èΩ,<NO¬±«~xǎ Åä£èü3Å
x}ö§k°Â
_yÍ ^ΔÉ] ,κ J/°iTê&1 'c<uΩ-
ÄD(G WÄC~y_ïõÄW PÔ1«ÎÜ†ç],κ;~Ì^üÑπ~≈~L~9OgflO~&ℰ≤ ¬≤ ØÔ§":
~ℰ!SGqèvo^ ú\,S>h<-*6ø‡%x'"|fiÓ#≈~my%~≥ñP<,fi Áj ÅÔ¿"Zù-
Ω·Õ-6ℰÿ{% „ΩÊó ,i π÷Áî°úO2çSÿ'O-
2Äflßi /@^"ΠK°ªPℰπ,úé^'3Σ~ö~ÔZÌ"Y¬ÿΩæY> Ω+eô/·<K£¿*÷~"≤û~
B ZøK~Qßÿüf,!ÒflÎzsS/]>ÈQ ü

Monoalphabetic Ciphers - Introduction

Limitation of Caesar Cipher

With only 25 possible keys, the Caesar cipher is far from secure and can be broken trivially by brute-force attack.

Solution: Arbitrary Substitution

A dramatic increase in key space can be achieved by allowing an arbitrary substitution where any plaintext letter can map to any ciphertext letter.

Permutation Definition

Before proceeding, we define the term **permutation**:

A **permutation** of a finite set of elements S is an **ordered sequence of all the elements of S** , with each element appearing exactly once.

Permutation - Mathematical Definition

Example with Small Set

If $S = \{a, b, c\}$, there are six permutations of S :

abc

acb

bac

bca

cab

cba

General Formula

In general, there are $n!$ permutations of a set of n elements because:

1st element

n ways

2nd element

$(n-1)$ ways

3rd element

$(n-2)$ ways

And so on...

...

 **Total permutations:** $n \times (n-1) \times (n-2) \times \dots \times 2 \times 1 = n!$ (n factorial)

Monoalphabetic Ciphers - Key Space

If the "cipher" line can be any permutation of the 26 alphabetic characters, then there are $26!$ possible keys.

Caesar Cipher

Key space: **25**

Brute-force: **Trivial**

Monoalphabetic

Key space: $26! \approx 4 \times 10^{26}$

Brute-force: **Impractical**

Comparison with DES

The monoalphabetic key space of $26!$ or greater than 4×10^{26} is:

10 orders of magnitude greater than the key space for DES

This would seem to **eliminate brute-force techniques** for cryptanalysis.

i Why "Monoalphabetic": Called monoalphabetic substitution cipher because a **single cipher alphabet** (mapping from plain alphabet to cipher alphabet) is used per message.

Monoalphabetic Ciphers - Cryptanalysis Example

Vulnerability to Frequency Analysis

If the cryptanalyst knows the nature of the plaintext (e.g., noncompressed English text), then the analyst can **exploit the regularities of the language**.

Ciphertext to be Solved

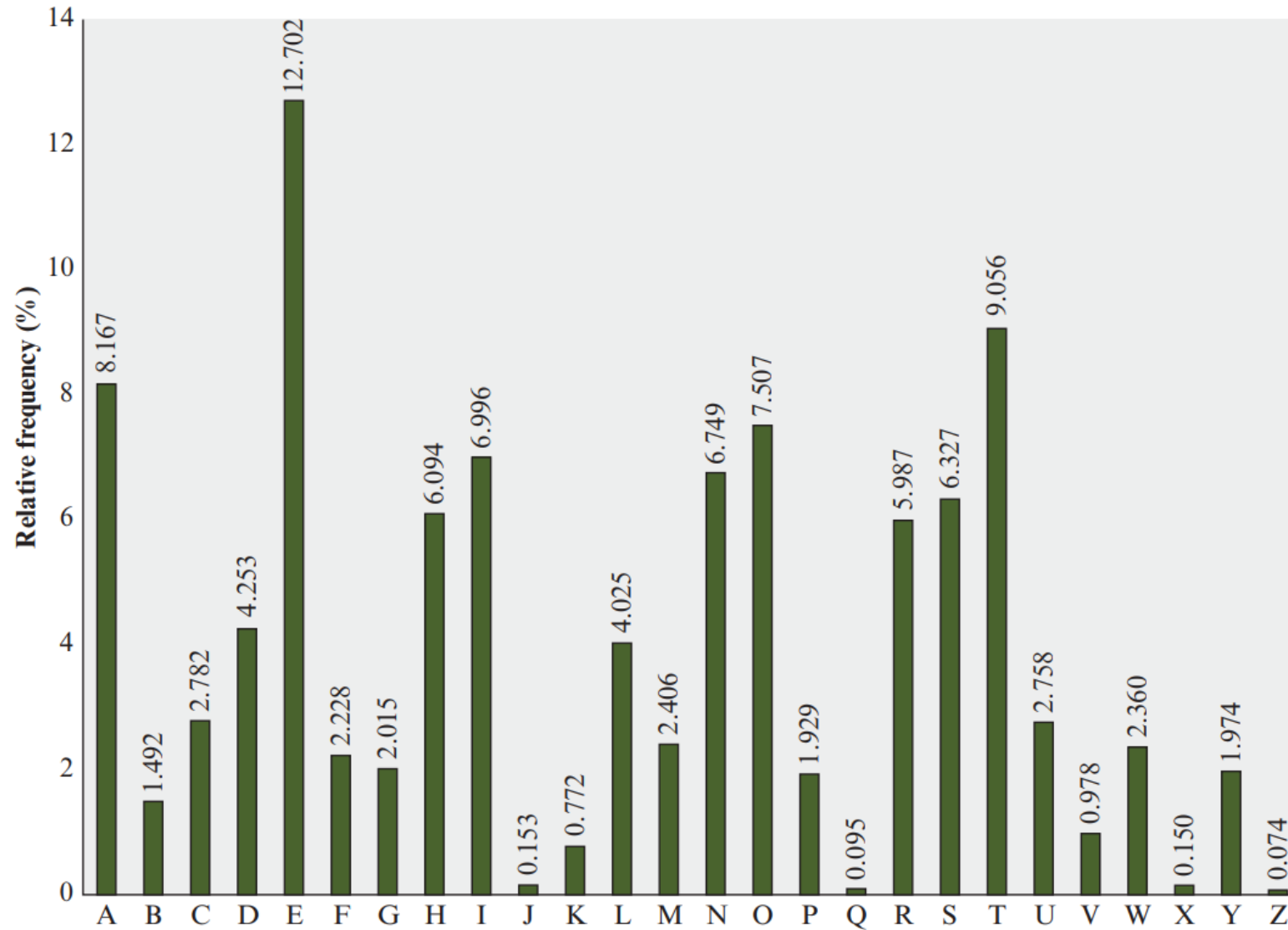
UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZVUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSXEPYE
POPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

This example is adapted from Sinkov's work on cryptanalysis techniques.

First Step: Frequency Analysis

As a first step, the **relative frequency of the letters** can be determined and compared to a standard frequency distribution for English text.

Relative Frequency of Letters in English Text



Frequency Analysis - Letter Counts

The relative frequencies of the letters in the ciphertext (in percentages) are as follows:

P 13.33%	Z 11.67%	S 8.33%	U 8.33%	O 7.50%
M 6.67%	H 5.83%	D 5.00%	E 5.00%	V 4.17%
X 4.17%	F 3.33%	W 3.33%	Q 2.50%	T 2.50%
A 1.67%	B 1.67%	G 1.67%	Y 1.67%	I 0.83%
J 0.83%	C 0.00%	K 0.00%	L 0.00%	N 0.00%
R 0.00%	Observation: If the message were long enough, frequency analysis alone might be sufficient, but this relatively short message requires additional techniques.			

Frequency Analysis - Initial Deductions

Comparing this breakdown with the previous Figure, we can make initial hypotheses about letter mappings:

High-Frequency Letters

Cipher P (13.33%) and Z (11.67%) likely correspond to plain e and t, though uncertain which is which.

Letters S, U, O, M, H of relatively high frequency probably correspond to plain letters from the set {a, h, i, n, o, r, s}.

Low-Frequency Letters

The letters with the **lowest frequencies** (namely, A, B, G, Y, I, J) are likely included in the set {b, j, k, q, v, x, z}.

 **Note:** Letters C, K, L, N, R have 0.00% frequency in this ciphertext.

→ **Next Steps:** There are several ways to proceed at this point, including making tentative assignments and looking for other regularities in the ciphertext.

Digram Analysis

Powerful Cryptanalytic Tool

A powerful tool is to look at the frequency of two-letter combinations, known as digrams.

Most Common English Digram

"th"

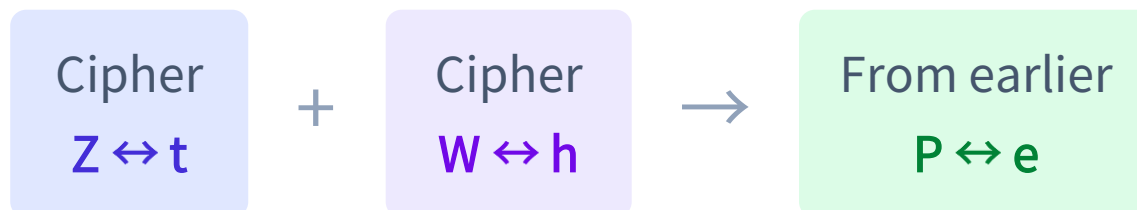
The most frequent two-letter combination in English text

Most Common Ciphertext Digram

"ZW"

Appears three times in the ciphertext

Making the Correspondence



Trigram Analysis and Progress

Sequence ZWP Analysis

Notice that the sequence **ZWP** appears in the ciphertext, and we can translate that sequence as **"the"**.

Most Frequent English Trigram

"the"

The most frequent three-letter combination in English

This correspondence seems to indicate that we are on the right track with our frequency analysis.

Sequence ZWSZ Analysis

Next, notice the sequence **ZWSZ** in the first line. We do not know that these four letters form a complete word, but if they do, it is of the form **th_t**.

If so, **S** equates with **a**.

Monoalphabetic Cipher - Complete Solution

Partial Decryption Progress

So far, with only four letters identified (Z→t, W→h, P→e, S→a), we already have quite a bit of the message:

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ

t a e e t e a that e e a a

Completing the Cryptanalysis

Continued analysis of frequencies plus trial and error should easily yield a solution from this point.

Complete Plaintext:
"it was disclosed yesterday that several informal but direct contacts have been made with political representatives of the viet cong in moscow"

💡 **Key Insight:** Even with a key space of 26!, monoalphabetic ciphers are vulnerable to frequency analysis because they preserve the statistical patterns of the original language.

Monoalphabetic Ciphers - Vulnerability

Fundamental Weakness

Monoalphabetic ciphers are easy to break because they reflect the frequency data of the original alphabet.

Countermeasure: Homophones

A countermeasure is to provide **multiple substitutes**, known as **homophones**, for a single letter.

Example with Letter 'e':

The letter **e** could be assigned a number of different cipher symbols, such as **16, 74, 35, and 21**, with each homophone assigned to a letter in rotation or randomly.

Proportional Assignment

If the number of symbols assigned to each letter is **proportional to the relative frequency** of that letter, then single-letter frequency information is completely obliterated.

Homophones - Limitations

Historical Perspective

The great mathematician **Carl Friedrich Gauss** believed that he had devised an **unbreakable cipher** using homophones.

However, Vulnerabilities Remain

Even with homophones, **each element of plaintext affects only one element of ciphertext**, and multiple-letter patterns (e.g., digram frequencies) still survive in the ciphertext.

Why Cryptanalysis Remains Straightforward

- ✗ Single-letter frequency may be obliterated, but digram and trigram patterns persist.
- ✗ The fundamental structure of the language is still preserved in the ciphertext.
- ✗ Statistical analysis of multi-letter combinations can still reveal the plaintext.

⚠ **Conclusion:** Making cryptanalysis relatively straightforward even with homophonic substitution.

Two Methods to Reduce Plaintext Structure

Two principal methods are used in substitution ciphers to lessen the extent to which the structure of the plaintext survives in the ciphertext:

1 Encrypt Multiple Letters

Encrypt **multiple letters of plaintext** at once rather than single letters.

Examples: Playfair Cipher (digrams), Hill Cipher (m-letter blocks)

2 Use Multiple Cipher Alphabets

Use **multiple cipher alphabets** and switch between them during encryption.

Examples: Vigenère Cipher, One-Time Pad

→ We briefly examine each approach in the following sections.

Playfair Cipher - Introduction

Best-Known Multiple-Letter Cipher

The **Playfair cipher** is the best-known multiple-letter encryption cipher, which treats **digrams in the plaintext as single units** and translates these units into ciphertext digrams.

Inventor

Actually invented by British scientist **Sir Charles Wheatstone** in **1854**.

Namesake

Bears the name of his friend **Baron Playfair of St. Andrews**, who championed the cipher at the British foreign office.

Literary Reference

The cipher was famously solved by **Lord Peter Wimsey** in Dorothy Sayers's mystery novel "**Have His Carcase**", which provides an absorbing account of a probable-word attack.

Playfair Cipher - Matrix Construction

The Playfair algorithm is based on the use of a 5×5 matrix of letters constructed using a keyword.

Matrix Construction Steps

- 1 Fill in the letters of the keyword (minus duplicates) from left to right and from top to bottom.
- 2 Fill in the remainder of the matrix with the remaining letters in alphabetic order.

Special Rule

The letters **I** and **J** count as one letter, allowing the 26-letter alphabet to fit in a 5×5 matrix (25 positions).

Example Keyword: "monarchy"

In this case, the keyword is **monarchy**. The matrix will be constructed by first placing M, O, N, A, R, C, H, Y (removing duplicates), then filling remaining cells with the rest of the alphabet (excluding J, which combines with I).

Playfair Cipher Matrix - Example

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Playfair Cipher - Encryption Rules

Plaintext is encrypted **two letters at a time**, according to the following rules:

1

Repeating Letters

Repeating plaintext letters that are in the same pair are separated with a **filler letter**, such as **x**, so that **balloon** would be treated as **balxloon**.

2

Same Row

Two plaintext letters that fall in the **same row** are each replaced by the letter to the **right**, with the first element of the row circularly following the last. For example, **ar** is encrypted as **RM**.

3

Same Column

Two plaintext letters that fall in the **same column** are each replaced by the letter **beneath**, with the top element of the column circularly following the last. For example, **mu** is encrypted as **CM**.

Playfair Cipher - Encryption Rules

4

Rectangle Rule

Otherwise, each plaintext letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter.

Examples:

Plaintext:

hs

→

BP

Plaintext:

ea

→

IM

(or JM)

Note: For the rectangle rule, the encipherer has some discretion when I/J is involved, as these letters share the same position in the matrix.

Playfair Cipher - Strength and Security

Great Advance Over Monoalphabetic Ciphers

The Playfair cipher is a **great advance** over simple monoalphabetic ciphers for several reasons.

Larger Symbol Set

Whereas there are only **26 letters**, there are:

$$26 \times 26 = 676$$

possible digrams

Making identification of individual digrams more difficult.

Flattened Frequency Distribution

The relative frequencies of individual letters exhibit a **much greater range** than that of digrams, making frequency analysis much more difficult.

Historical Usage

For these reasons, the Playfair cipher was for a long time **considered unbreakable**. It was used as the **standard field system** by the **British Army in World War I** and still enjoyed considerable use by the **U.S. Army and other Allied forces during World War II**.

Playfair Cipher - Vulnerability

Despite Confidence, Still Breakable

Despite this level of confidence in its security, the Playfair cipher is relatively easy to break, because it still leaves much of the structure of the plaintext language intact.



Cryptanalysis Requirements

~100-200

letters of ciphertext

are generally sufficient to break the cipher



Remaining Weakness

The cipher still preserves much of the language structure, including:

- Some digram frequency patterns
- Word structure information
- Language-specific patterns



Key Lesson: No cipher that preserves significant language structure can be considered truly secure, regardless of its key space size.

Frequency Distribution Analysis

Figure ahead provides a powerful way to visualize the **effectiveness of different ciphers** at masking frequency information.

Plaintext Distribution

The line labeled plaintext plots a typical frequency distribution of the 26 alphabetic characters in ordinary text.

Normalization: occurrences of each letter divided by occurrences of most frequent letter (e).

Playfair Distribution

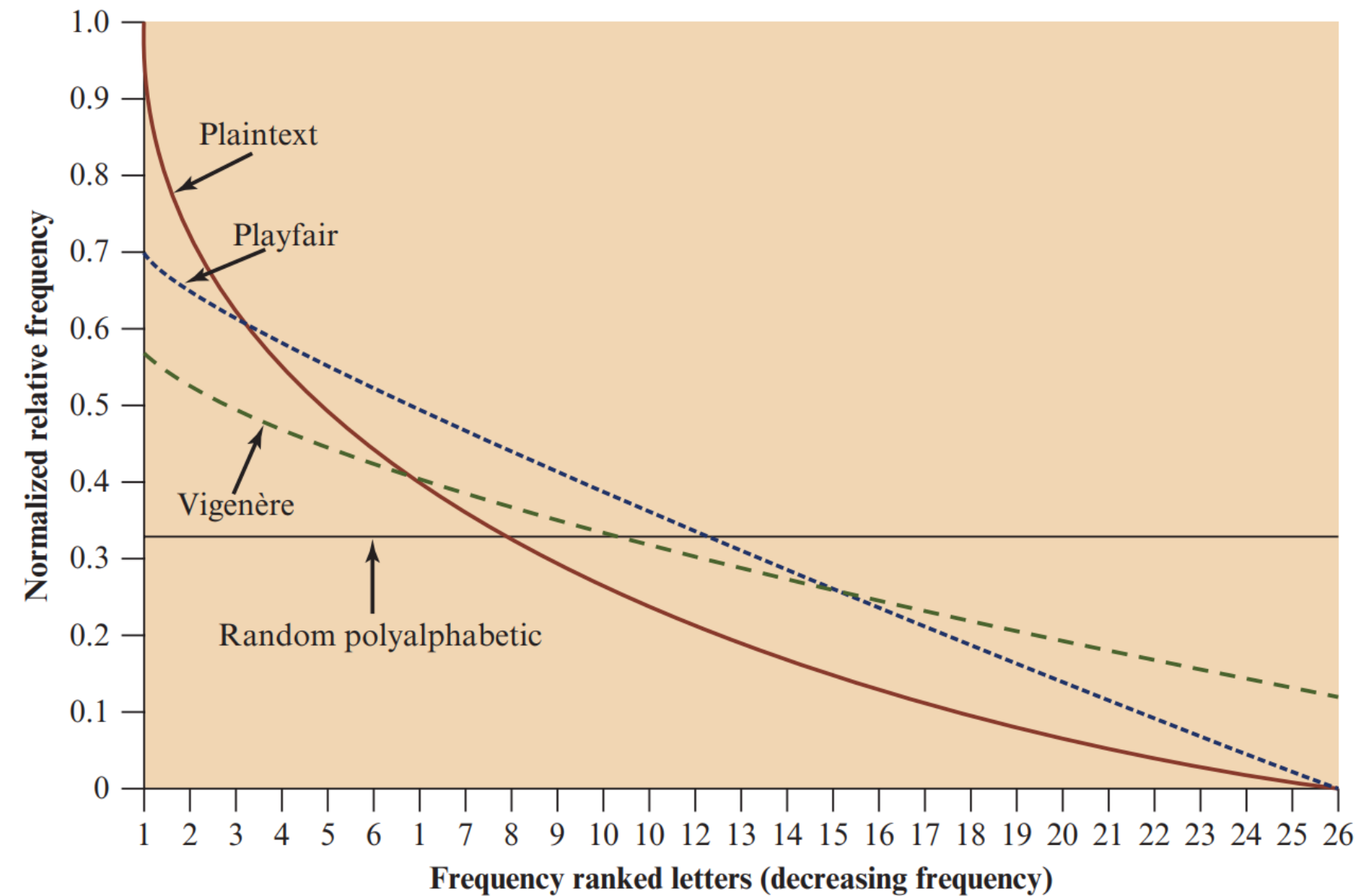
The Playfair cipher has a **flatter distribution** than does plaintext, but nevertheless, it reveals plenty of structure for a cryptanalyst to work with.

Shows improvement over simple substitution but still vulnerable.

Ideal Distribution

If frequency distribution information were **totally concealed** in the encryption process, the ciphertext plot of frequencies would be **flat**, and cryptanalysis using ciphertext only would be **effectively impossible**.

Relative Frequency of Occurrence of Letters



Hill Cipher - Introduction

Another Multiletter Cipher

Another interesting multiletter cipher is the **Hill cipher**, developed by the mathematician **Lester Hill in 1929**.



Educational Value

Illustrates an **important point about cryptanalysis** that will be useful later on.



Mathematical Foundation

More difficult to understand than other ciphers but demonstrates **mathematical principles** in encryption.



Note: Understanding this subsection provides valuable insights into linear algebra applications in cryptography.

Hill Cipher - Linear Algebra Concepts

Concepts from Linear Algebra

Before describing the Hill cipher, let us briefly review some terminology from linear algebra. In this discussion, we are concerned with **matrix arithmetic modulo 26**.

Matrix Inverse Definition

We define the **inverse** M^{-1} of a square matrix M by the equation:

$$M(M^{-1}) = M^{-1}M = I$$

where I is the identity matrix

Identity Matrix

I is a square matrix that is all zeros except for ones along the main diagonal from upper left to lower right.

$$I = \begin{bmatrix} 1, & 0, & 0, \\ 0, & 1, & 0, \\ 0, & 0, & 1 \end{bmatrix}$$

Hill Cipher - Matrix Inverse Example

The inverse of a matrix does not always exist, but when it does, it satisfies the preceding equation.

Matrix A

$$\begin{bmatrix} 5 & 17 \\ 8 & 3 \end{bmatrix}$$

Inverse $A^{-1} \bmod 26$

$$\begin{bmatrix} 9 & 1 \\ 2 & 15 \end{bmatrix}$$

Product AA^{-1}

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Verification Calculation

$$\begin{aligned} AA^{-1} &= [(5 \times 9) + (8 \times 1), (5 \times 1) + (8 \times 15), \\ &\quad (17 \times 9) + (3 \times 1), (17 \times 1) + (3 \times 15)] \\ &= [[53, 130], [156, 79]] \bmod 26 \\ &= [[1, 0], [0, 1]] \checkmark \end{aligned}$$

Hill Cipher - Determinant

To explain how the inverse of a matrix is computed, we begin with the concept of **determinant**.

2×2 Matrix Determinant

For matrix $[[k_{11}, k_{12}], [k_{21}, k_{22}]]$:

$$\det = k_{11}k_{22} - k_{12}k_{21}$$

3×3 Matrix Determinant

More complex formula:

$$\begin{aligned} \det = & k_{11}k_{22}k_{33} + k_{21}k_{32}k_{13} + k_{31}k_{12}k_{23} \\ & - k_{31}k_{22}k_{13} - k_{21}k_{12}k_{33} - k_{11}k_{32}k_{23} \end{aligned}$$

Inverse Computation Formula

If a square matrix A has a **nonzero determinant**, then the inverse is computed as:

$$[A^{-1}]_{ij} = (\det A)^{-1}(-1)^{i+j}(D_{ji})$$

where (D_{ji}) is the subdeterminant formed by deleting the j th row and i th column

Hill Cipher - Determinant Example

Continuing Our Example

$$\begin{aligned}\text{For } A &= \begin{bmatrix} 5 & 17 \\ 8 & 3 \end{bmatrix}: \\ \det &= (5 \times 3) - (8 \times 17) \\ &= 15 - 136 \\ &= -121 \bmod 26 \\ &= 9\end{aligned}$$

Multiplicative Inverse

We can show that $9^{-1} \bmod 26 = 3$, because:

$$9 \times 3 = 27 \bmod 26 = 1 \checkmark$$

Computing the Inverse

$$A^{-1} \bmod 26 = 3 \times \begin{bmatrix} 3 & -17 \\ -8 & 5 \end{bmatrix} = \begin{bmatrix} 9 & -51 \\ -24 & 15 \end{bmatrix} \bmod 26 = \begin{bmatrix} 9 & 1 \\ 2 & 15 \end{bmatrix} \checkmark$$

Hill Cipher - Algorithm

The Hill Algorithm

This encryption algorithm takes m successive plaintext letters and substitutes for them m ciphertext letters.

Substitution by Linear Equations

The substitution is determined by m linear equations in which each character is assigned a numerical value ($a=0, b=1, \dots, z=25$).

For $m = 3$:

$$c_1 = (k_{11}p_1 + k_{21}p_2 + k_{31}p_3) \bmod 26$$

$$c_2 = (k_{12}p_1 + k_{22}p_2 + k_{32}p_3) \bmod 26$$

$$c_3 = (k_{13}p_1 + k_{23}p_2 + k_{33}p_3) \bmod 26$$

 **Note:** The value of m determines the block size. Larger m values provide more security but require larger key matrices.

Hill Cipher - Matrix Form

This can be expressed in terms of **row vectors** and **matrices**:

Matrix Equation

$$(c_1 \ c_2 \ c_3) = (p_1 \ p_2 \ p_3) \times K \text{ mod } 26$$

$$\mathbf{C} = \mathbf{PK} \text{ mod } 26$$

Variables

C and P: Row vectors of length m

K: $m \times m$ matrix (encryption key)

Operations: Performed mod 26

Alternative Convention

Some cryptography books express plaintext and ciphertext as **column vectors**, placing the column vector after the matrix. Sage uses row vectors, so we adopt that convention.

Hill Cipher - Encryption Example

Example: Consider the plaintext "paymoremoney" and use the encryption key K.

Encryption Key K

$$\begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$$

First Three Letters

Plaintext: "pay"

Vector: (15 0 24)

p=15, a=0, y=24

Encryption Calculation

$$\begin{aligned} (15 \ 0 \ 24) \times K &= (303 \ 303 \ 531) \bmod 26 \\ &= (17 \ 17 \ 11) \\ &= \text{RRL} \end{aligned}$$

Continuing in this fashion, the ciphertext for the entire plaintext is RRLMWBKASPDH.

Hill Cipher - Decryption

Decryption requires using the inverse of the matrix K .

Computing the Inverse

$$\det K = 23$$

$$(\det K)^{-1} \bmod 26 = 17$$

$$K^{-1} = \begin{bmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{bmatrix}$$

Verification

$$K \times K^{-1} = \begin{bmatrix} 443 & 858 & 494 \\ 442 & 495 & 52 \\ 442 & 780 & 365 \end{bmatrix}$$

$$\bmod 26 =$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \checkmark$$

✓ **Result:** It is easily seen that if the matrix K^{-1} is applied to the ciphertext, then the plaintext is recovered.

Hill Cipher - General Form

In general terms, the Hill system can be expressed as:

Encryption

$$C = E(K, P) = PK \bmod 26$$

Decryption

$$\begin{aligned} P &= D(K, C) = CK^{-1} \bmod 26 \\ &= PKK^{-1} = P \end{aligned}$$

Strength of Hill Cipher

- ✓ As with Playfair, the strength of the Hill cipher is that it **completely hides single-letter frequencies**.
- ✓ With Hill, the use of a **larger matrix hides more frequency information**.
- ✓ A **3×3 Hill cipher** hides not only single-letter but also **two-letter frequency information**.

Hill Cipher - Known Plaintext Attack

Vulnerability to Known Plaintext

Although the Hill cipher is strong against a ciphertext-only attack, it is easily broken with a known plaintext attack.

Attack Requirements

For an $m \times m$ Hill cipher, suppose we have m plaintext-ciphertext pairs, each of length m .

Label the pairs:

$$P_j = (p_{1j} \ p_{2j} \ \dots \ p_{mj})$$

$$C_j = (c_{1j} \ c_{2j} \ \dots \ c_{mj})$$

Such that $C_j = P_j K$ for $1 \leq j \leq m$

Matrix Equation Solution

Define two $m \times m$ matrices $X = (p_{ij})$ and $Y = (c_{ij})$. Then we can form the matrix equation:

$$Y = XK$$

If X has an inverse:

$$K = X^{-1}Y$$

Hill Cipher - Cryptanalysis Example

Example: Suppose that the plaintext "hillcipher" is encrypted using a 2×2 Hill cipher to yield the ciphertext HCRZSSXNSP.

Known Pairs

$$(7 \ 8)K \bmod 26 = (7 \ 2)$$

$$h(7)i(8) \rightarrow H(7)C(2)$$

$$(11 \ 11)K \bmod 26 = (17 \ 25)$$

$$l(11)l(11) \rightarrow R(17)Z(25)$$

Matrix Equation

$$\begin{bmatrix} 7 & 2 \\ 17 & 25 \end{bmatrix} = \begin{bmatrix} 7 & 11 \\ 11 & 11 \end{bmatrix} \times K \bmod 26$$

$$Y = XK$$

Solving for K

$$X^{-1} = \begin{bmatrix} 25 & 1 \\ 22 & 23 \end{bmatrix}$$

$$\begin{aligned} K &= X^{-1}Y \\ &= \begin{bmatrix} 3 & 8 \\ 2 & 5 \end{bmatrix} \end{aligned}$$

This result is **verified by testing the remaining plaintext-ciphertext pairs.**

Polyalphabetic Ciphers - Introduction

Improving on Monoalphabetic Technique

Another way to improve on the simple monoalphabetic technique is to use **different monoalphabetic substitutions as one proceeds through the plaintext message**.

General Name and Features

The general name for this approach is **polyalphabetic substitution cipher**. All these techniques have the following features in common:

1 Multiple Rules

A set of related monoalphabetic substitution rules is used.

2 Key Determines Rule

A key determines which particular rule is chosen for a given transformation.

→ We will examine: Vigenère Cipher, Vernam Cipher, and One-Time Pad.

Vigenère Cipher - Introduction

Best Known Polyalphabetic Cipher

The Vigenère cipher is the best known, and one of the simplest, polyalphabetic ciphers.

Cipher Structure

In this scheme, the set of related monoalphabetic substitution rules consists of the **26 Caesar ciphers with shifts of 0 through 25**.

Each cipher is denoted by a **key letter**, which is the ciphertext letter that substitutes for the plaintext letter **a**. Thus, a Caesar cipher with a shift of 3 is denoted by the key value 3.

 **Helper Tool:** To aid in understanding and use, a matrix known as the Vigenère tableau is often used (discussed in supplementary material).

Vigenère Cipher - Mathematical Formulation

We can express the Vigenère cipher in the following manner:

Sequence Definitions

Plaintext: $P = p_0, p_1, p_2, \dots, p_{n-1}$

Key: $K = k_0, k_1, k_2, \dots, k_{m-1}$ (where $m < n$)

Ciphertext: $C = C_0, C_1, C_2, \dots, C_{n-1}$

Encryption Process

The key letters are **repeated** as needed to match plaintext length:

$$C = (p_0 + k_0) \bmod 26, (p_1 + k_1) \bmod 26, \dots, (p_{m-1} + k_{m-1}) \bmod 26, \\ (p_m + k_0) \bmod 26, (p_{m+1} + k_1) \bmod 26, \dots$$

Vigenère Cipher - Encryption Example

To encrypt a message, a key is needed that is as long as the message. Usually, the key is a repeating keyword.

Example

Key: deceptive deceptive deceptive

Plaintext: wearediscoveredsaveyourself

Ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

How It Works

Position 1:

$w(22) + d(3)$

$= Z(25)$

Position 2:

$e(4) + e(4)$

$= I(8)$

Position 3:

$a(0) + c(2)$

$= C(2)$

And so on, with the key repeating every 9 letters.

Vigenère Cipher - Numerical Example

Expressed numerically, we have the following result:

Numerical Demonstration

Position	Key	Plaintext	Calculation	Ciphertext
1	d(3)	w(22)	$(22+3) \bmod 26$	Z(25)
2	e(4)	e(4)	$(4+4) \bmod 26$	I(8)
3	c(2)	a(0)	$(0+2) \bmod 26$	C(2)
4	e(4)	r(17)	$(17+4) \bmod 26$	V(21)
5	p(15)	e(4)	$(4+15) \bmod 26$	T(19)

This process continues through the entire message with the key repeating every 9 letters.

Vigenère Cipher - Strength

Primary Strength

The strength of this cipher is that there are **multiple ciphertext letters for each plaintext letter**, one for each unique letter of the keyword.

Frequency Obscuration

Thus, the **letter frequency information is obscured**, making simple frequency analysis ineffective.



Remaining Weakness

However, **not all knowledge of the plaintext structure is lost**. Statistical patterns still exist.

Frequency Distribution Evidence

Figure 3.6 shows the frequency distribution for a Vigenère cipher with a keyword of length 9:

An improvement is achieved over the Playfair cipher, but considerable frequency information remains, making cryptanalysis possible.

Vigenère Cipher - Cryptanalysis Method

It is instructive to sketch a method of breaking this cipher, because the method reveals some of the **mathematical principles that apply in cryptanalysis**.

First Step: Determine Cipher Type

First, suppose that the opponent believes that the ciphertext was encrypted using either **monoalphabetic substitution** or a **Vigenère cipher**.

Simple test: If monoalphabetic substitution is used, then the statistical properties of the ciphertext should be the same as that of the language of the plaintext.

Expected Frequency Distribution

Referring to Figure 3.5, there should be:

~12.7%

One cipher letter

~9.06%

Another letter

etc.

Following distribution

If correspondence is close, we can assume **monoalphabetic substitution**.

Vigenère Cipher - Determining Keyword Length

If, on the other hand, a Vigenère cipher is suspected, then progress depends on **determining the length of the keyword**.

Key Insight

The important insight that leads to a solution: If two identical sequences of plaintext letters occur at a distance that is an integer multiple of the keyword length, they will generate identical ciphertext sequences.

Example from Our Ciphertext

In the foregoing example, two instances of the sequence **"red"** are separated by **nine character positions**.

r encrypted with
key letter e

e encrypted with
key letter p

d encrypted with
key letter t

Thus, in both cases, the **ciphertext sequence** is VTW.

Vigenère Cipher - Keyword Length Analysis

An analyst looking at only the ciphertext would detect the repeated sequences and make assumptions about keyword length.

Detecting Repeated Sequences

Detect repeated sequence **VTW** at a displacement of **9**.

Make the assumption that the keyword is either **3 or 9 letters** in length (factors of 9).

Statistical Considerations

- ❓ The appearance of VTW twice **could be by chance** and may not reflect identical plaintext letters encrypted with identical key letters.
- ✅ However, if the message is long enough, there will be **a number of such repeated ciphertext sequences**.

💡 **Method:** By looking for common factors in the displacements of the various sequences, the analyst should be able to make a good guess of the keyword length.

Vigenère Cipher - Solving Individual Ciphers

Important Insight

Solution of the cipher now depends on an important insight: If the keyword length is m , then the cipher, in effect, **consists of m monoalphabetic substitution ciphers**.

Example with Keyword DECEPTIVE

With the keyword DECEPTIVE (9 letters):

Positions

1, 10, 19, ...

encrypted with same cipher

Positions

2, 11, 20, ...

encrypted with same cipher

And so on...

through position 9

9 separate monoalphabetic ciphers

Attack Strategy

Thus, we can use the **known frequency characteristics of the plaintext language** to attack each of the monoalphabetic ciphers separately, breaking the overall cipher piece by piece.

Autokey System

Eliminating Periodicity

The periodic nature of the keyword can be eliminated by using a **nonrepeating keyword** that is as long as the message itself.

Vigenère's Autokey System

Vigenère proposed what is referred to as an **autokey system**, in which a keyword is concatenated with the plaintext itself to provide a running key.

Key: deceptivewearediscoveredsav

Plaintext: wearediscoveredsaveyourself

Ciphertext: ZICVTWQNGKZEIIGASXSTSLVWLA

i How it works: After the initial keyword "deceptive" is used, the plaintext itself becomes the continuation of the key, creating a key as long as the message.

Autokey System - Vulnerability

Still Vulnerable

Even this scheme is **vulnerable to cryptanalysis**.

The Problem

Because the key and the plaintext **share the same frequency distribution of letters**, a statistical technique can be applied.

Example Analysis

e enciphered by e: Can be expected to occur with a frequency of $(0.127)^2 \approx 0.016$

t enciphered by t: Would occur only about **half as often**

 **Conclusion:** These regularities can be exploited to achieve successful cryptanalysis, even of the autokey system.

Vernam Cipher - Introduction

Ultimate Defense

The ultimate defense against such cryptanalysis is to choose a keyword that is **as long as the plaintext and has no statistical relationship to it.**

Inventor

Such a system was introduced by an **AT&T engineer named Gilbert Vernam in 1918.**

Binary Operation

His system works on **binary data (bits)** rather than letters, making it suitable for telegraph and digital systems.

Mathematical Expression

The system can be expressed succinctly as follows:

$$c_i = p_i \oplus k_i$$

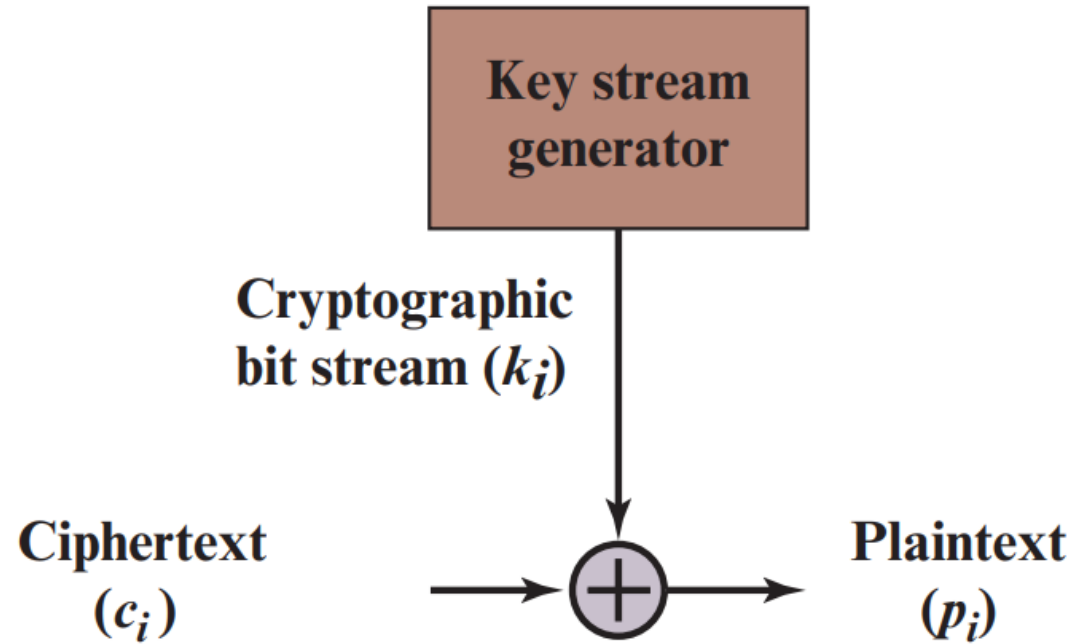
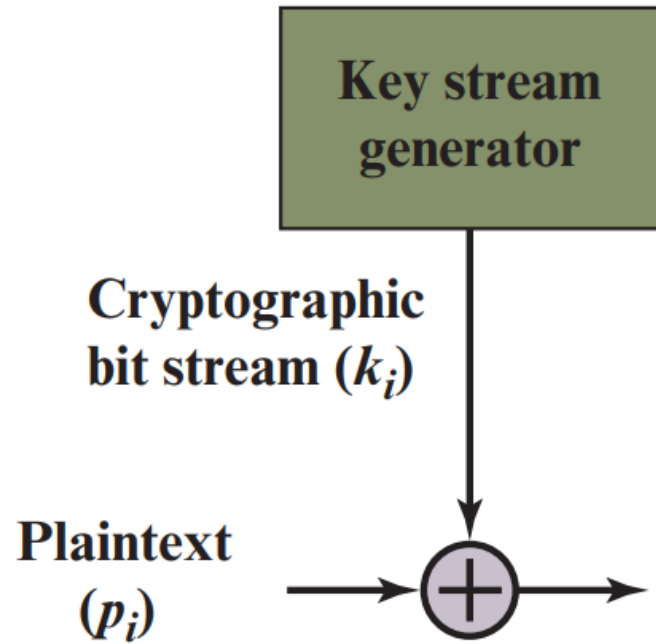
p_i = ith plaintext bit

k_i = ith key bit

c_i = ith ciphertext bit

$\oplus$ = exclusive-OR (XOR) operation

Vernam Cipher



Vernam Cipher - Operation

Compare this with Equation (3.3) for the Vigenère cipher. In essence, each plaintext character is encrypted with a different mechanism.

Encryption

$$c_i = p_i \oplus k_i$$

Ciphertext generated by XOR of plaintext and key

Decryption

$$p_i = c_i \oplus k_i$$

Same XOR operation recovers plaintext

Key Construction

The essence of this technique is the **means of construction of the key**.

Vernam proposed the use of a **running loop of tape** that eventually repeated the key, so that in fact the system worked with a **very long but repeating keyword**.

Vernam Cipher - Vulnerability

Still Breakable

Although such a scheme, with a long key, presents **formidable cryptanalytic difficulties**, it can be broken with:



Sufficient Ciphertext

Large amounts of ciphertext enable statistical analysis



Known Plaintext

Use of known or probable plaintext sequences



Both

Combination of both approaches

⚠ Key Weakness: The repeating nature of the key (even if very long) creates patterns that can be exploited by cryptanalysts.

One-Time Pad - Introduction

Ultimate Security Improvement

An Army Signal Corp officer, Joseph Mauborgne, proposed an improvement to the Vernam cipher that yields the **ultimate** in security.

Key Requirements

- 1 Use a random key that is as long as the message, so that the key need not be repeated.
- 2 The key is to be used to encrypt and decrypt a **single message**, and then is discarded.
- 3 Each new message requires a **new key of the same length** as the new message.

The One-Time Pad

Such a scheme, known as a **one-time pad**, is unbreakable.

One-Time Pad - Security

Why It's Unbreakable

It produces **random output** that bears no statistical relationship to the **plaintext**. Because the ciphertext contains no information whatsoever about the plaintext, there is simply no way to break the code.

Mathematical Proof

Given any plaintext of equal length to the ciphertext, there is a key that produces that plaintext.

Therefore, if you did an exhaustive search of all possible keys, you would end up with **many legible plaintexts**, with no way of knowing which was the intended plaintext.

Therefore, the code is unbreakable.

 **Security Foundation:** The security of the one-time pad is entirely due to the randomness of the key. If the stream of characters that constitute the key is truly random, then the stream of characters that constitute the ciphertext will be truly random.

One-Time Pad - Example

Suppose that we are using a Vigenère scheme with 27 characters (including space) but with a one-time key that is as long as the message.

Ciphertext

ANKYODKYUREPFJBYOJDSPLEIYUNOFDOIUERFPLUYTS

Decryption 1

Ciphertext: ANKYODKYUREPFJBYOJDSPLEIYUNOFDOIUERFPLUYTS

Key: pxlmvmsydofoyrvzwc tlnbnevcgdupahfzzlmnyih

Plaintext: mr mustard with the candlestick in the hall

Decryption 2

Ciphertext: ANKYODKYUREPFJBYOJDSPLEIYUNOFDOIUERFPLUYTS

Key: pftgpmiydgaxgoufhkllmhsqdgogtewbqfgyovuhwt

Plaintext: miss scarlet with the knife in the library

❓ **The Dilemma:** Two plausible plaintexts are produced. How is the cryptanalyst to decide which is the correct decryption? If the actual key was produced in a truly random fashion, then the cryptanalyst cannot say that one of these two keys is more likely than the other.

One-Time Pad - Randomness

Security Due to Randomness

The security of the one-time pad is **entirely due to the randomness of the key**.

Randomness Implications



If the stream of characters that constitute the key is **truly random**, then the stream of characters that constitute the ciphertext will be **truly random**.



Thus, there are **no patterns or regularities** that a cryptanalyst can use to attack the ciphertext.

Theoretical Perfection

In theory, we **need look no further for a cipher**. The one-time pad offers complete security but, in practice, has two fundamental difficulties.

One-Time Pad - Practical Difficulties

Two Fundamental Difficulties in Practice

The one-time pad offers complete security but, in practice, has **two fundamental difficulties**:

1 Key Generation

There is the **practical problem of making large quantities of random keys**.

Any heavily used system might require **millions of random characters on a regular basis**. Supplying truly random characters in this volume is a significant task.

2 Key Distribution

Even more daunting is the **problem of key distribution and protection**.

For every message to be sent, a **key of equal length is needed by both sender and receiver**. Thus, a mammoth key distribution problem exists.

⚠ Consequence: Because of these difficulties, the one-time pad is of limited utility and is useful primarily for low-bandwidth channels requiring very high security.

One-Time Pad - Utility

Because of these difficulties, the one-time pad is of **limited utility** and is useful primarily for **low-bandwidth channels requiring very high security**.

Perfect Secrecy

The one-time pad is the **only cryptosystem** that exhibits what is referred to as **perfect secrecy**.

Further Reading

This concept is explored in **Appendix B** of the textbook for those interested in the mathematical foundations.

Appropriate Use Cases

- ✓ Diplomatic communications with highest security requirements
- ✓ Military command channels with limited message volume
- ✓ Situations where keys can be securely distributed in advance

 **Key Takeaway:** While theoretically perfect, the one-time pad's practical limitations prevent its widespread use in modern communication systems.

Transposition Techniques - Introduction

All the techniques examined so far involve the **substitution** of a ciphertext symbol for a plaintext symbol. A very different kind of mapping is achieved by performing some sort of **permutation** on the plaintext letters.

Transposition Cipher Definition

This technique is referred to as a **transposition cipher**.



Rail Fence

Simplest technique writing diagonals



Columnar

Write rows, read columns



Double

Multiple transposition stages

Rail Fence Technique

The simplest such cipher is the **rail fence technique**, in which the plaintext is written down as a sequence of diagonals and then read off as a sequence of rows.

Example

To encipher the message "meet me after the toga party" with a rail fence of **depth 2**, we write the following:

Row 1: m e m a t r h t g p r y

Row 2: e t e f e t e o a a t

Encrypted message:

MEMATRHTGPRYETEFETEOAAT

 **Security Note:** This sort of thing would be **trivial to cryptanalyze** due to its simple structure and preserved letter frequencies.

Columnar Transposition

A more complex scheme is to write the message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns.

Example with Key 4312567

Plaintext written in rows:

Key: 4 3 1 2 5 6 7

Row 1: a t t a c k p

Row 2: o s t p o n e

Row 3: d u n t i l t

Row 4: w o a m x y z

Reading by key order:

Start with column 1 (labeled 3), then column 4 (labeled 2), then column 2 (labeled 1), etc.

TTNAAPTMTSUOAODWCOIXKNLYPETZ

 The order of the columns becomes the key to the algorithm. In this example, the key is 4312567.

Columnar Transposition - Cryptanalysis

Recognizing Transposition

A pure transposition cipher is easily recognized because it has the **same letter frequencies as the original plaintext**.

Cryptanalysis Approach

For the type of columnar transposition just shown, cryptanalysis is fairly straightforward and involves:

- 1 Laying out the ciphertext in a **matrix** with appropriate dimensions.
- 2 Playing around with **column positions** to find meaningful text.
- 3 Using **digram and trigram frequency tables** to guide the search.

 **Key Insight:** Since letter frequencies are preserved, the cryptanalyst knows they are dealing with a transposition cipher and can use language statistics to guide the column reordering.

Double Transposition

The transposition cipher can be made significantly more secure by performing **more than one stage of transposition**.

Result

The result is a **more complex permutation that is not easily reconstructed**.

Example: Reencrypting with Same Algorithm

If the foregoing message is reencrypted using the same algorithm with key **4312567**:

Input (from first encryption):

ttnaapt
mtsua o
dwcoixk
nlypetz

Output (after second encryption):

NSCYAUOPTTWLTMDNAOIEPAXTTOKZ

Double Transposition - Position Analysis

To visualize the result of this double transposition, designate the letters in the original plaintext message by the numbers designating their position.

Position Tracking (28 letters)

Original sequence:

01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28

After first transposition:

03 10 17 24 04 11 18 25 02 09 16 23 01 08 15 22 05 12 19 26 06 13 20 27 07 14 21 28

(has a somewhat regular structure)

After second transposition:

17 09 05 27 24 16 12 07 10 02 22 20 03 25 15 13 04 23 19 14 11 01 26 21 18 08 06 28

(much less structured permutation)

 **Security Improvement:** This is a much less structured permutation and is much more difficult to cryptanalyze than a single transposition.



Acknowledgement

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Thank you for your attention

CS464 - Data and Network Security

Week 03 - Classical Encryption Techniques